

Abstract Linear Algebra

Prof. Lanier

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Pick up a handout. Try those warm-up problems.

Next up:

- Please fill out the Mid-quarter course check-in.
- Quizzes 6 & 7 today
- Problem session today, 3-5pm, Ryerson 177.
- HW₄ due tonight / Saturday.
- HW₅ distributed today. ***Change:*** Due a week from Monday, not a week from today.
- Midterm exam review on Sunday in Cobb 115, 3:00-4:30pm
- Midterm exam on Monday
- HW₄ debrief is Monday afternoon, 3:30-4:30, in Eckhart 207A

Big results about determinants

For $n \times n$ matrices A and B , $\det(AB) = \det(A) \det(B)$ and $\det(A^{-1}) = \det(A)^{-1}$ (Thm. 3.3.5)

($\det$ is a homomorphism)

$$\det A = \det A^T$$

(Cor: row operations affect $\det$ the same way column operations do.)

You know the quadratic formula. What about the linear systems formula?

Why does cofactor expansion work?

Warm-up 1

Find the determinant of each of the following matrices:

$$\det \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 8 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} =$$

$$\det \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -3 \end{pmatrix} =$$

$$\det \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} =$$

$$\det \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -24 & 0 & 0 & -3 \\ 48 & 0 & 9 & 0 \end{pmatrix} =$$

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$$\det A \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 8 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} =$$

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$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -24 & 0 & 0 & -3 \\ 48 & 0 & 9 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -3 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 8 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -3 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -3 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 8 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 8 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -24 & 0 & 0 & -3 \\ 48 & 0 & 9 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -3 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 8 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -3 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -3 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 8 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 8 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

-3
-1
1
-3
-1
-3
1
-1
1

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -24 & 0 & 0 & -3 \\ 48 & 0 & 9 & 0 \end{pmatrix} = A \begin{matrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} & \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 8 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} & \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -3 \end{pmatrix} & \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} & \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -3 \end{pmatrix} & \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 8 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} & \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} & \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 8 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \\ -3 & -1 & 1 & -3 & -1 & -3 & 1 & -1 & 1 \end{matrix}$$

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -24 & 0 & 0 & -3 \\ 48 & 0 & 9 & 0 \end{pmatrix} = A \begin{matrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 8 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} & \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -3 \end{pmatrix} & \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} & \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -3 \end{pmatrix} & \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 8 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} & \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} & \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 8 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \\ 3 & 1 & -3 & -1 & -3 & 1 & -1 & 1 \end{matrix}$$

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$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -24 & 0 & 0 & -3 \\ 48 & 0 & 9 & 0 \end{pmatrix} = A \begin{matrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -3 \end{pmatrix} & \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 8 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} & \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} & \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 8 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \\ 9 & -3 & 1 & -1 & 1 \end{matrix}$$

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$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -24 & 0 & 0 & -3 \\ 48 & 0 & 9 & 0 \end{pmatrix} = A \begin{matrix} -27 & -1 & 1 \end{matrix}$$

$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$

$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 8 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -24 & 0 & 0 & -3 \\ 48 & 0 & 9 & 0 \end{pmatrix} = A_{27}^1 \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 8 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -24 & 0 & 0 & -3 \\ 48 & 0 & 9 & 0 \end{pmatrix} =$$

A

27

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -24 & 0 & 0 & -3 \\ 48 & 0 & 9 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -24 & 0 & 0 & -3 \\ 48 & 0 & 9 & 0 \end{pmatrix}$$

27

Every invertible matrix is the product of elementary matrices.

But $\det(E_i)$ is easily checked, and so is $\det(AE_i) = \det(A) \det(E_i)$

This follows:

For $n \times n$ matrices A and B , $\det(AB) = \det(A) \det(B)$ and $\det(A^{-1}) = \det(A)^{-1}$ (Thm. 3.3.5)

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Pf. $A = E_1 \dots E_m$ $B = F_1 \dots F_n$ $AB = E_1 \dots E_m F_1 \dots F_n$

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$$\det A = \det E_1 \dots \det E_m \qquad \det B = \det F_1 \dots \det F_n$$

$$\det(AB) = \det E_1 \dots \det E_m \det F_1 \dots \det F_n \quad \blacksquare$$

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Pf. $AA^{-1} = I$

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Pf.

$$AA^{-1} = I$$

$$\det(A) \det(A^{-1}) = \det(I)$$

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Pf.

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$$\det(A) \det(A^{-1}) = \det(I)$$

$$\det(A) \det(A^{-1}) = 1$$

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Pf.

$$AA^{-1} = I$$

$$\det(A) \det(A^{-1}) = \det(I)$$

$$\det(A) \det(A^{-1}) = 1$$

$$\det(A^{-1}) = 1 / \det(A) \quad \blacksquare$$

$$\det A = \det A^T$$

Check that it's true for the E_i
Recall how transposes and products interact:
$$(AB)^T = B^T A^T$$

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Check that it's true for the E_i

Recall how transposes and products interact:

$$(\hat{A}B)^T = B^T \hat{A}^T$$

Do a column operation to A ,
take the transpose. $=$ Take the transpose of A .
do the corresponding
row operation.

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$$\begin{aligned} \det A &= \det(E_1 E_2 \dots E_n) \\ &= \det(E_1) \det(E_2) \dots \det(E_n) \\ &= \det(E_n) \dots \det(E_2) \det(E_1) \\ &= \det(E_n^T) \dots \det(E_2^T) \det(E_1^T) \\ &= \det(E_n^T \dots E_2^T E_1^T) \\ &= \det((E_1 E_2 \dots E_n)^T) \\ &= \det A^T \end{aligned}$$

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Corollary: row operations affect det the same way column operations do.

quadratic formula

$$ax^2 + bx + c = 0 \Rightarrow$$

$$x^2 + \frac{b}{a}x + \frac{c}{a} = 0 \Rightarrow$$

$$x^2 + \frac{b}{a}x + \frac{b^2}{4a^2} = +\frac{b^2}{4a^2} - \frac{c}{a} \Rightarrow \dots$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\begin{array}{l} ax + cy = e \\ bx + dy = f \end{array} \Rightarrow$$

quadratic formula

$$ax^2 + bx + c = 0 \Rightarrow x^2 + \frac{b}{a}x + \frac{c}{a} = 0 \Rightarrow x^2 + \frac{b}{a}x + \frac{b^2}{4a^2} = +\frac{b^2}{4a^2} - \frac{c}{a} \Rightarrow \dots$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

linear systems formula (Cramer's rule)

$$\begin{array}{l} ax + cy = e \\ bx + dy = f \end{array} \Rightarrow x = \frac{\begin{vmatrix} e & c \\ f & d \end{vmatrix}}{\begin{vmatrix} a & c \\ b & d \end{vmatrix}} \quad y = \frac{\begin{vmatrix} a & e \\ b & f \end{vmatrix}}{\begin{vmatrix} a & c \\ b & d \end{vmatrix}}$$

quadratic formula

$$ax^2 + bx + c = 0 \Rightarrow x^2 + \frac{b}{a}x + \frac{c}{a} = 0 \Rightarrow x^2 + \frac{b}{a}x + \frac{b^2}{4a^2} = +\frac{b^2}{4a^2} - \frac{c}{a} \Rightarrow \dots$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

linear systems formula (Cramer's rule)

$$\begin{array}{l} ax + cy = e \\ bx + dy = f \end{array} \Rightarrow x = \frac{\begin{vmatrix} e & c \\ f & d \end{vmatrix}}{\begin{vmatrix} a & c \\ b & d \end{vmatrix}} \quad y = \frac{\begin{vmatrix} a & e \\ b & f \end{vmatrix}}{\begin{vmatrix} a & c \\ b & d \end{vmatrix}}$$

$$3x + 2y = 13$$

$$5x - 2y = 11$$

Assume at least one of a, b is non-zero. WLOG, take it to be a .

$$ax + cy = e$$

$$bx + dy = f$$

Assume at least one of a, b is non-zero. WLOG, take it to be a .

$$ax + cy = e$$

$$bx + dy = f$$

$$abx + cby = eb$$

$$abx + ady = af$$

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$$abx + ady = af$$

$$abx + cby = eb$$

$$(ad - bc)y = af - eb$$

Assume at least one of a, b is non-zero. WLOG, take it to be a .

Assume that $ad-bc$ is non-zero.

$$ax + cy = e$$

$$bx + dy = f$$

$$abx + cby = eb$$

$$abx + ady = af$$

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$$(ad - bc)y = af - eb$$

Assume at least one of a, b is non-zero. WLOG, take it to be a .

Assume that $ad-bc$ is non-zero.

$$ax + cy = e$$

$$bx + dy = f$$

$$abx + cby = eb$$

$$y = \frac{af - eb}{ad - bc}$$

$$abx + cby = eb$$

$$abx + ady = af$$

$$abx + cby = eb$$

$$(ad - bc)y = af - eb$$

Assume at least one of a, b is non-zero. WLOG, take it to be a .

Assume that $ad-bc$ is non-zero.

$$ax + cy = e$$

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$$abx + cby = eb$$

$$y = \frac{af - eb}{ad - bc}$$

$$abx + cby = eb$$

$$abx + ady = af$$

$$abx = eb - bc \frac{af - eb}{ad - bc}$$

$$y = \frac{af - eb}{ad - bc}$$

$$abx + cby = eb$$

$$(ad - bc)y = af - eb$$

$$x = \frac{ed - cf}{ad - bc}$$

$$y = \frac{af - eb}{ad - bc}$$

Assume at least one of a, b is non-zero. WLOG, take it to be a .

Assume that $ad-bc$ is non-zero.

$$ax + cy = e$$

$$bx + dy = f$$

$$abx + cby = eb$$

$$y = \frac{af - eb}{ad - bc}$$

$$abx + cby = eb$$

$$abx + ady = af$$

$$abx = eb - bc \frac{af - eb}{ad - bc}$$

$$y = \frac{af - eb}{ad - bc}$$

$$abx + cby = eb$$

$$(ad - bc)y = af - eb$$

$$x = \frac{\begin{vmatrix} e & c \\ f & d \end{vmatrix}}{\begin{vmatrix} a & c \\ b & d \end{vmatrix}}$$

$$y = \frac{\begin{vmatrix} a & e \\ b & f \end{vmatrix}}{\begin{vmatrix} a & c \\ b & d \end{vmatrix}}$$

$$x = \frac{ed - cf}{ad - bc}$$

$$y = \frac{af - eb}{ad - bc}$$

Proof of cofactor expansion, 3×3 case

$$\begin{vmatrix} a & d & g \\ b & e & h \\ c & f & i \end{vmatrix} = \begin{vmatrix} a & d & g \\ 0 & e & h \\ 0 & f & i \end{vmatrix} + \begin{vmatrix} 0 & d & g \\ b & e & h \\ 0 & f & i \end{vmatrix} + \begin{vmatrix} 0 & d & g \\ 0 & e & h \\ c & f & i \end{vmatrix}$$

Proof of cofactor expansion, 3×3 case

$$\begin{vmatrix} a & d & g \\ b & e & h \\ c & f & i \end{vmatrix} = \begin{vmatrix} a & d & g \\ 0 & e & h \\ 0 & f & i \end{vmatrix} + \begin{vmatrix} 0 & d & g \\ b & e & h \\ 0 & f & i \end{vmatrix} + \begin{vmatrix} 0 & d & g \\ 0 & e & h \\ c & f & i \end{vmatrix}$$
$$= \begin{vmatrix} a & 0 & 0 \\ 0 & e & h \\ 0 & f & i \end{vmatrix} + \begin{vmatrix} 0 & d & g \\ b & 0 & 0 \\ 0 & f & i \end{vmatrix} + \begin{vmatrix} 0 & d & g \\ 0 & e & h \\ c & 0 & 0 \end{vmatrix}$$

Proof of cofactor expansion, 3×3 case

$$\begin{aligned} \begin{vmatrix} a & d & g \\ b & e & h \\ c & f & i \end{vmatrix} &= \begin{vmatrix} a & d & g \\ 0 & e & h \\ 0 & f & i \end{vmatrix} + \begin{vmatrix} 0 & d & g \\ b & e & h \\ 0 & f & i \end{vmatrix} + \begin{vmatrix} 0 & d & g \\ 0 & e & h \\ c & f & i \end{vmatrix} \\ &= \begin{vmatrix} a & 0 & 0 \\ 0 & e & h \\ 0 & f & i \end{vmatrix} + \begin{vmatrix} 0 & d & g \\ b & 0 & 0 \\ 0 & f & i \end{vmatrix} + \begin{vmatrix} 0 & d & g \\ 0 & e & h \\ c & 0 & 0 \end{vmatrix} \\ &= \begin{vmatrix} a & 0 & 0 \\ 0 & e & h \\ 0 & f & i \end{vmatrix} - \begin{vmatrix} b & 0 & 0 \\ 0 & d & g \\ 0 & f & i \end{vmatrix} + \begin{vmatrix} c & 0 & 0 \\ 0 & d & g \\ 0 & e & h \end{vmatrix} \end{aligned}$$

Proof of cofactor expansion, 3×3 case

$$\begin{aligned}
 \begin{vmatrix} a & d & g \\ b & e & h \\ c & f & i \end{vmatrix} &= \begin{vmatrix} a & d & g \\ 0 & e & h \\ 0 & f & i \end{vmatrix} + \begin{vmatrix} 0 & d & g \\ b & e & h \\ 0 & f & i \end{vmatrix} + \begin{vmatrix} 0 & d & g \\ 0 & e & h \\ c & f & i \end{vmatrix} \\
 &= \begin{vmatrix} a & 0 & 0 \\ 0 & e & h \\ 0 & f & i \end{vmatrix} + \begin{vmatrix} 0 & d & g \\ b & 0 & 0 \\ 0 & f & i \end{vmatrix} + \begin{vmatrix} 0 & d & g \\ 0 & e & h \\ c & 0 & 0 \end{vmatrix} \\
 &= \begin{vmatrix} a & 0 & 0 \\ 0 & e & h \\ 0 & f & i \end{vmatrix} - \begin{vmatrix} b & 0 & 0 \\ 0 & d & g \\ 0 & f & i \end{vmatrix} + \begin{vmatrix} c & 0 & 0 \\ 0 & d & g \\ 0 & e & h \end{vmatrix} \\
 &= a \begin{vmatrix} 1 & 0 & 0 \\ 0 & e & h \\ 0 & f & i \end{vmatrix} - b \begin{vmatrix} 1 & 0 & 0 \\ 0 & d & g \\ 0 & f & i \end{vmatrix} + c \begin{vmatrix} 1 & 0 & 0 \\ 0 & d & g \\ 0 & e & h \end{vmatrix}
 \end{aligned}$$

Proof of cofactor expansion, 3×3 case

$$\begin{aligned}
 \begin{vmatrix} a & d & g \\ b & e & h \\ c & f & i \end{vmatrix} &= \begin{vmatrix} a & d & g \\ 0 & e & h \\ 0 & f & i \end{vmatrix} + \begin{vmatrix} 0 & d & g \\ b & e & h \\ 0 & f & i \end{vmatrix} + \begin{vmatrix} 0 & d & g \\ 0 & e & h \\ c & f & i \end{vmatrix} \\
 &= \begin{vmatrix} a & 0 & 0 \\ 0 & e & h \\ 0 & f & i \end{vmatrix} + \begin{vmatrix} 0 & d & g \\ b & 0 & 0 \\ 0 & f & i \end{vmatrix} + \begin{vmatrix} 0 & d & g \\ 0 & e & h \\ c & 0 & 0 \end{vmatrix} \\
 &= \begin{vmatrix} a & 0 & 0 \\ 0 & e & h \\ 0 & f & i \end{vmatrix} - \begin{vmatrix} b & 0 & 0 \\ 0 & d & g \\ 0 & f & i \end{vmatrix} + \begin{vmatrix} c & 0 & 0 \\ 0 & d & g \\ 0 & e & h \end{vmatrix} \\
 &= a \begin{vmatrix} 1 & 0 & 0 \\ 0 & e & h \\ 0 & f & i \end{vmatrix} - b \begin{vmatrix} 1 & 0 & 0 \\ 0 & d & g \\ 0 & f & i \end{vmatrix} + c \begin{vmatrix} 1 & 0 & 0 \\ 0 & d & g \\ 0 & e & h \end{vmatrix} = a \begin{vmatrix} e & h \\ f & i \end{vmatrix} - b \begin{vmatrix} d & g \\ f & i \end{vmatrix} + c \begin{vmatrix} d & g \\ e & h \end{vmatrix}
 \end{aligned}$$

Next up:

- Please fill out the Mid-quarter course check-in.
- Quizzes 6 & 7 today
- Problem session today, 3-5pm, Ryerson 177.
- HW₄ due tonight / Saturday.
- HW₅ distributed today. ***Change:*** Due a week from Monday, not a week from today.
- Midterm exam review on Sunday in Cobb 115, 3:00-4:30pm
- Midterm exam on Monday
- HW₄ debrief is Monday afternoon, 3:30-4:30, in Eckhart 207A